

FACULTY OF ENGINEERING
DIPLOMA ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY



Introduction to Machine Learning (BE103711)

QUESTION BANK mid exam

5 MARKS QUESTIONS (15 Questions)

UNIT 1: Introduction to Machine Learning:

1. Explain the **Overview of Human Learning and Machine Learning**. Compare both with suitable examples.
2. Describe the **Types of Machine Learning**:
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learningwith examples.
3. Explain the **Applications of Machine Learning** in different fields such as healthcare, finance, education, e-commerce, and transportation.
4. Discuss the **Tools and Technologies used in Machine Learning**, including their features and uses.
5. Compare **Human Learning and Machine Learning** based on:
 - Learning process
 - Decision-making
 - Adaptability
 - SpeedAccuracy

UNIT 2: Preparing to Model:

1. Explain the **Machine Learning process (activities)** with a neat diagram.
2. Describe the **Types and Structures of Data** used in Machine Learning with suitable examples.
3. Explain **Data Quality** and discuss different **Data Remediation techniques**.
4. Explain the **Data Preprocessing** techniques used in Machine Learning, including:
 - Data Cleaning
 - Data Transformation
 - Data Integration
 - Data Reduction
5. Explain the following feature selection techniques with suitable examples:
 - Dimensionality Reduction
 - Feature Subset Selection

UNIT 3: Modeling and Evaluation:

1. Explain the **process of selecting a Machine Learning model**. Differentiate between **Predictive** and **Descriptive** models with suitable examples.
2. Describe the **steps involved in training a supervised learning model** with a neat diagram.
3. Explain **Model Representation** and **Model Interpretability** with suitable examples.

4. Discuss the **different methods used to evaluate the performance of a Machine Learning model**.
5. Explain the **techniques used to improve the performance of a Machine Learning model**, such as:
 - Feature Selection
 - Hyperparameter Tuning
 - Cross-Validation
 - Handling Overfitting and Underfitting

MULTIPLE CHOICE QUESTIONS (30 Questions)

UNIT 1: Introduction to Machine Learning:

1. Machine Learning is a subset of:

- A) Data Mining
- B) Artificial Intelligence
- C) Cloud Computing
- D) Cyber Security

✓ **Answer: B**

2. Which type of Machine Learning uses labeled data for training?

- A) Unsupervised Learning
- B) Reinforcement Learning
- C) Supervised Learning
- D) Deep Learning

✓ **Answer: C**

3. Which of the following is a common application of Machine Learning?

- A) Spam Detection
- B) Weather Forecasting
- C) Recommendation Systems
- D) All of the Above

✓ **Answer: D**

4. Which of the following is a popular programming language used for Machine Learning?

- A) Python
- B) HTML
- C) CSS
- D) SQL

✓ **Answer: A**

5. Which type of Machine Learning learns by interacting with an environment using rewards and penalties?

- A) Supervised Learning
- B) Unsupervised Learning
- C) Reinforcement Learning
- D) Semi-Supervised Learning

✓ **Answer: C**

UNIT 2: Preparing to Model:

1. Which of the following is the first step in a Machine Learning project?

- A) Model Deployment
- B) Data Collection
- C) Model Evaluation
- D) Prediction

✓ **Answer: B**

2. Which data quality issue occurs when some values are unavailable?

- A) Duplicate Data
- B) Missing Data
- C) Noise
- D) Redundant Data

✓ **Answer: B**

3. Dimensionality Reduction is mainly used to:

- A) Increase the number of features
- B) Reduce the number of features while preserving important information
- C) Increase training time
- D) Create duplicate features

✓ **Answer: B**

4. Feature Subset Selection is used to:

- A) Remove irrelevant and redundant features
- B) Increase data size
- C) Create new datasets
- D) Label the data automatically

✓ **Answer: A**

5. Which of the following is an important step in Data Preprocessing?

- A) Data Cleaning
- B) Feature Selection
- C) Data Transformation
- D) All of the Above

✓ **Answer: D**

UNIT 3: Modeling and Evaluation:

1. Which type of model is mainly used to predict future outcomes?

- A) Descriptive Model
- B) Predictive Model
- C) Clustering Model
- D) Association Model

✓ **Answer: B**

2. In supervised learning, a model is trained using:

- A) Unlabeled Data
- B) Random Data
- C) Labeled Data
- D) Incomplete Data

☒ **Answer: C**

3. Model Interpretability refers to:

- A) Increasing the size of the dataset
- B) Understanding how a model makes predictions
- C) Reducing the number of features
- D) Improving hardware performance

☒ **Answer: B**

4. Which of the following is used to evaluate the performance of a Machine Learning model?

- A) Accuracy
- B) Precision
- C) Recall
- D) All of the Above

☒ **Answer: D**

5. Which technique is commonly used to improve the performance of a Machine Learning model?

- A) Feature Selection
- B) Hyperparameter Tuning
- C) Cross-Validation
- D) All of the Above

☒ **Answer: D**